

Lab 1.5 - Basic Assembly using MARS Simulator

Objective

The purpose of this lab is to

- get familiar with running basic commands using the MARS simulator
- run a simple arithmetic commands on the simulator
- get familiar with the MARS IDE layout

Lab Learning Outcomes

By completion of the lab, the student should be able to

1. Know how to execute basic MIPS commands on the MARS simulator
2. Write a simple MIPS program and run it in the MARS simulator

Lab Exercises

The following tasks should be completed individually within the duration of the lab

1. Write a program in MIPS that adds two numbers stored in registers \$s1 and \$s4 and stores the result in register \$t3. Use the `addi` command to load the numbers into registers directly. Assemble the MIPS code, execute it and verify using the register pane if the operation was successfully completed.
2. Repeat problem #1 to find the difference of two numbers using the `sub` command.
3. Modify your program to read two numbers from the user, then add them and display the result to the user.
[Note: You will need to use the `move` command to move the numbers between registers \$a and \$v0 and the other registers. Use the help menu in the MARS IDE to learn how to use the move command.]
4. Modify your program to display appropriate messages while reading values from the user and while displaying the results.
5. Write a program in MIPS to evaluate the expression $14423 + 1248 - (33452 - 4560) + 2(6873)$ using only the MIPS instructions you have learnt in the theory class.